

From Learning AI → Building AI

WELCOME TO THE AI ERA

Presented by Ali Raafat

From Learning AI → Building AI



AI is already making decisions around you.

Then show 4-6 examples visually:

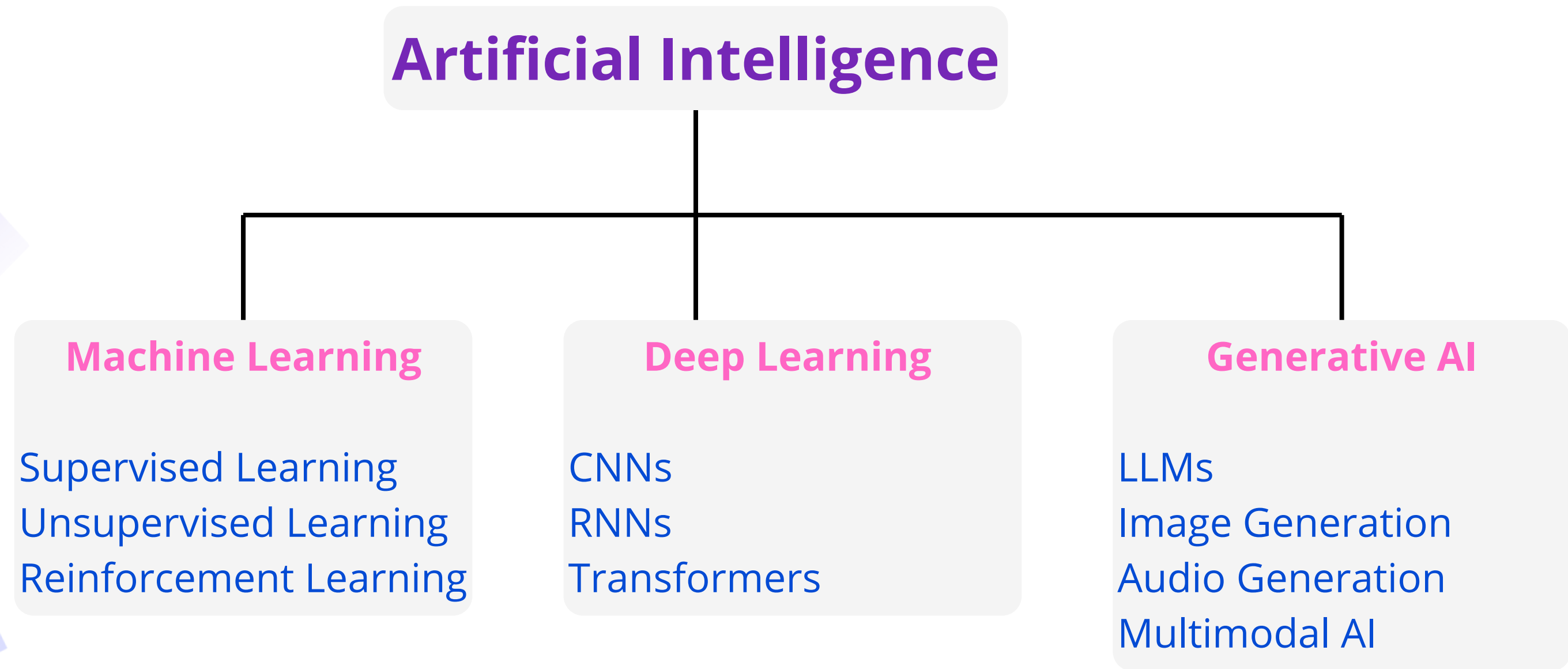
- Netflix → What should you watch?
- Google Maps → Which route should you take?
- Instagram/TikTok → What should you see next?
- Gmail → Is this spam?
- Banks → Is this transaction suspicious?
- ChatGPT → How should information be generated?

So... What Exactly Is AI?

- A programming language?
- A model?
- A robot?
- ChatGPT?
- Machine Learning?
- Deep Learning?
- All of them?



AI Is an Ecosystem



AI is the field.

ML, DL, and Generative AI are different approaches within that field.

WHAT DOES AN AI ENGINEER ACTUALLY DO

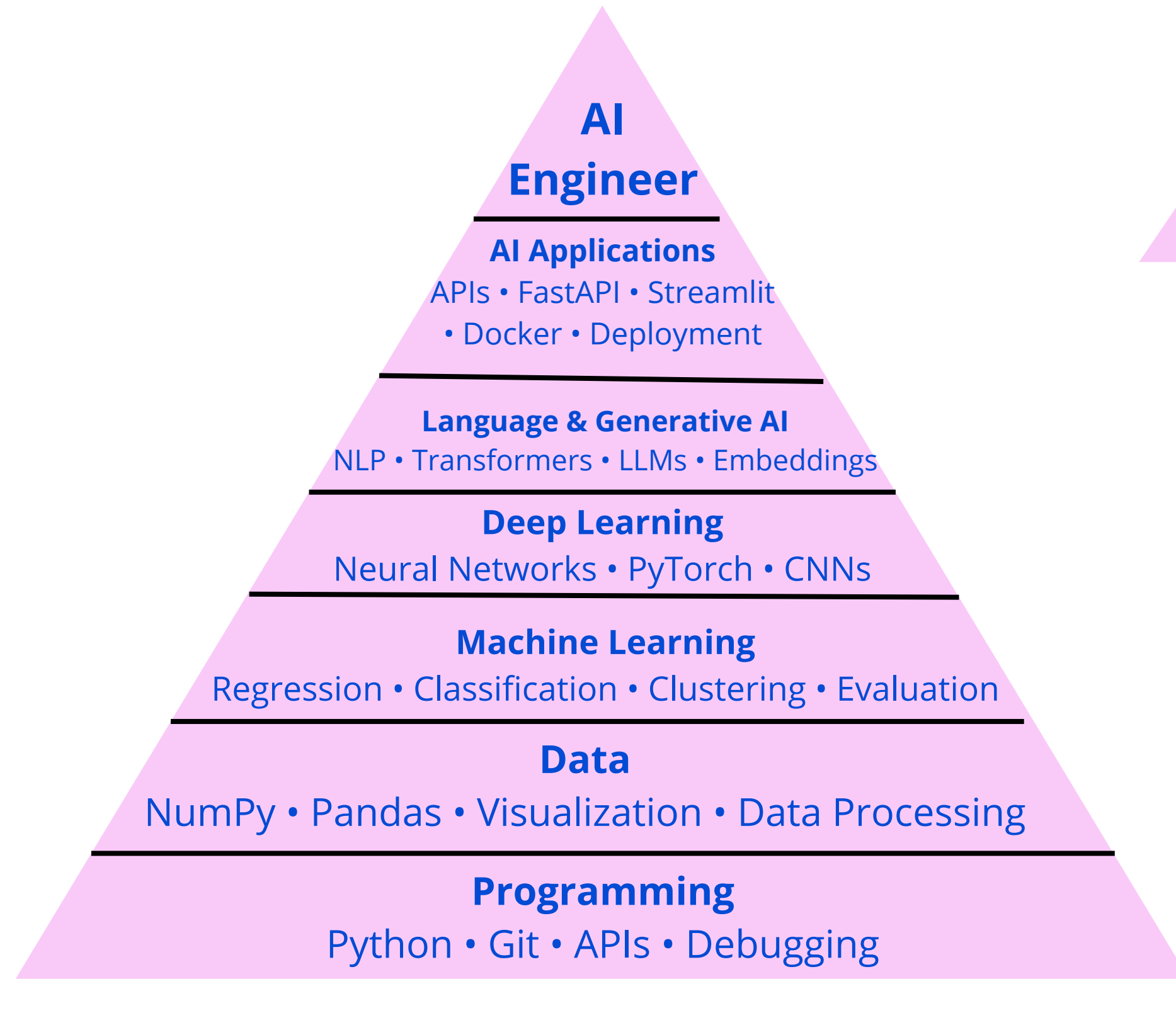
Problem → Data → Exploration → Preprocessing



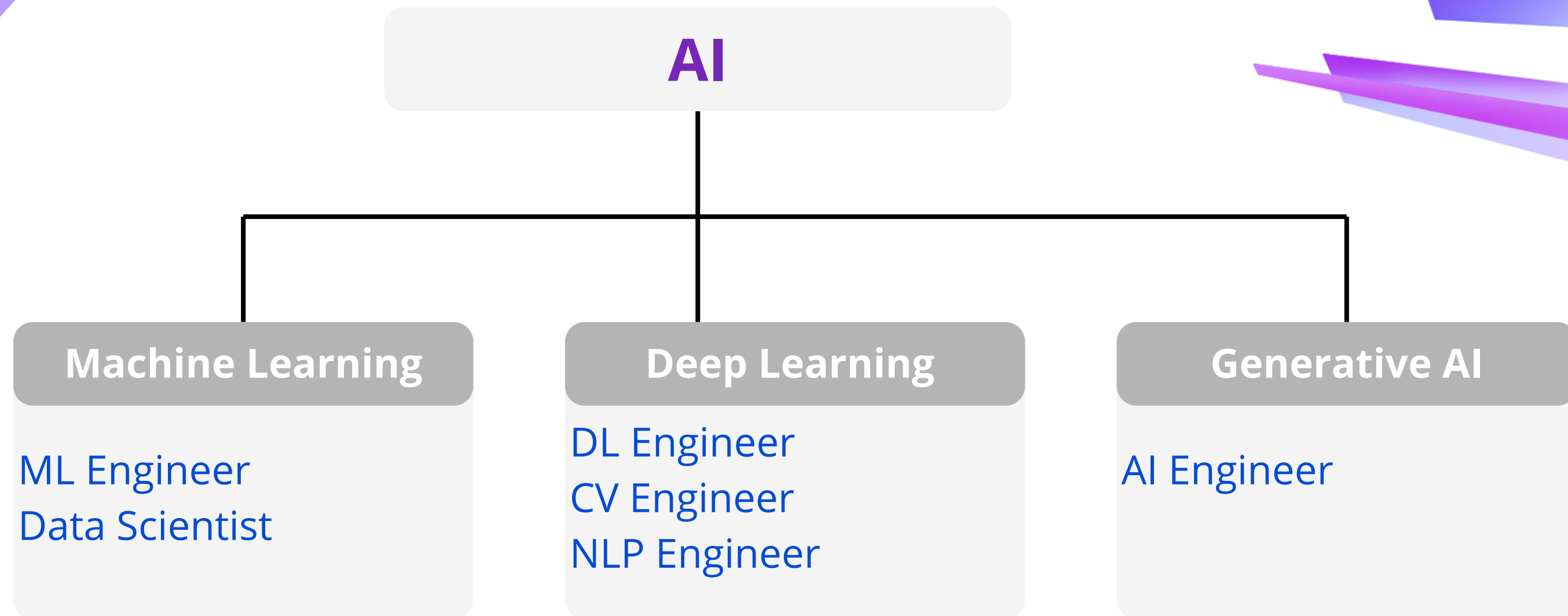
Features / Representations → Model → Evaluation

Application → Deployment → Monitoring →
Improvement

The Real AI Stack



AI Job Market



AI As Your Assistant

Bad Workflow

Problem → Ask AI → Copy → Run
→ Error → Ask AI again → Copy

TriNode Workflow

Understand → Think → Design →
Ask AI → Verify → Test → Improve

TriNode Roadmap



MODULE 0

Engineering Mindset

MODULE 1

Python for AI

MODULE 2

Machine Learning

MODULE 3

Deep Learning

MODULE 4

NLP

MODULE 5

LLMs & Generative AI

MODULE 6

AI Applications & Deployment



Programming Language

Different Problems → Different Languages

Language	Common Uses
Python	AI, Data, Automation, Backend
C / C++	Systems, Embedded
Java	Backend, Android
JavaScript	Web Applications
C#	.NET, Game Development
Dart	Cross-Platform Application Development

Why Python

Not because:

“Python is the easiest language.”

And not because:

“Everyone uses it.”

We chose Python because it gives us a powerful combination of:

Simplicity

- Easy-to-read syntax

Productivity

- Build ideas quickly

Ecosystem

- Huge collection of AI & data libraries

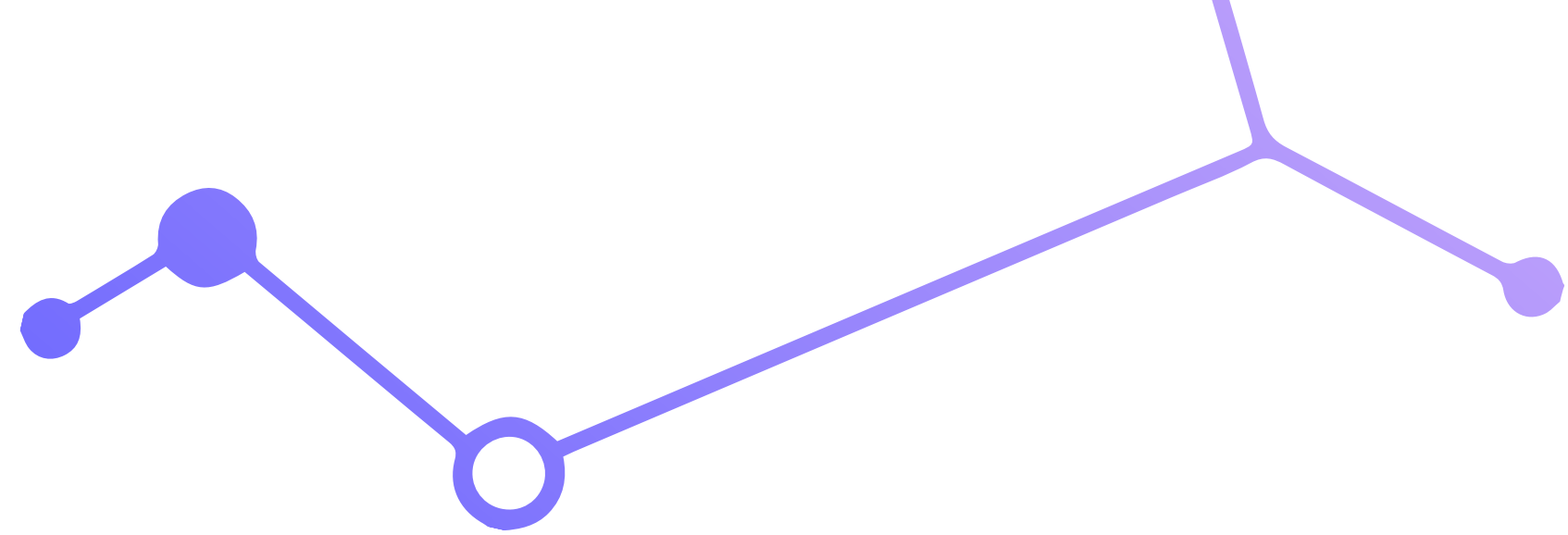
Community

- Massive developer and research community

AI Adoption

- Widely used across modern AI workflows





Thank You

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Best Wishes TriNode